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An Improved Data Anomaly Detection Method Based on Isolation Forest

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Abstract—**An improved data anomaly detection method SA-iForest is proposed to solve the problem of low accuracy, poor execution efficiency and generalization ability of data anomalies detection algorithm based on isolated forest. Based on the idea of selective integration, the precision and the difference value are taken as the criterion, and the simulated annealing algorithm is used to select the isolation tree with high abnormality detection and differentity to optimize the forest. At the same time, the excess detection precision is small and the difference is small Isolation tree improves the forest construction process of isolated forests, which improves the efficiency of the algorithm and improves the efficiency of the algorithm. The method of data anomaly detection based on SA-iForest is compared with the traditional Isolation Forest algorithm and LOF algorithm, and the accuracy and efficiency of the data are verified by the standard simulation data set. There is a significant improvement.**

Keywords- Isolation Forest; Outlier detection; SA-iForest; Simulated annealing.

I. INTRODUCTION

Data anomaly detection as one of the important tasks of data mining [1], usually find some interesting and valuable model, so much attention in the industry, so the emergence of a lot of good anomaly detection methods. Data anomaly detection and evaluation methods can be divided into: based on distance [2], based on density [3], based on model [4-6] and so on. Among them, Liu et al. [6] proposed isolation forest (Isolation Forest) algorithm by calculating the length of the data to be tested for abnormal detection. Compared with other anomaly detection algorithms, this algorithm reduces the masking and flooding effects of anomalies, reduces the computational overhead, and has the linear time complexity. The scalability of the high-dimensional data is better, but the algorithm introduces the random factors, Resulting in low accuracy and poor stability and other issues [7,8]. [8] the iForestASD algorithm is proposed, which uses the sliding window to deal with the flow data and considers the concept of drift. The algorithm can effectively detect the abnormal instances of the stream data. Yu et al. [9] proposed a combination of iForest and LOF, which improved the accuracy and stability of the algorithm, but there was a problem of reduced efficiency. [10] an iForest algorithm based on relative quality improvement is proposed. The algorithm is based on the global ranking of path length and local ranking. It solves the problem that local anomaly is masked by normal data clusters with similar density, but increases computational overhead.

Compared with other anomaly detection algorithms, Isolation Forest does not use distance or density to detect anomalies, avoiding large computations based on distance and density methods. The use of exception data and the characteristics of different, so that they are quickly separated from the normal data, with a low linear time complexity. Although the Isolation Forest algorithm has achieved good results in anomaly detection, there are still some shortcomings:

a. Isolation Forest algorithm for the accuracy of the data set detection and the number of iTree related to the construction of large-scale iTree need to spend more memory space, while leading to greater computational complexity.

b. With the increase in the number of iTree, iTree differences between the smaller and smaller, resulting in reduced generalization performance. At the same time, the increase in the number of iTree will lead to a large number of poor performance iTree, and the accuracy of its abnormal detection uneven.

c. Isolation Forest algorithm is based on a single computing node design, the size of the data processed by the maximum memory capacity limit, the processing of massive data is more difficult, and the introduction of random factors, resulting in low accuracy and poor stability and other issues.

Aiming at the above problem, this paper proposes a data anomaly detection method based on SA-iForest. The method uses Simulated Annealing algorithm to optimize iForest, which enhances the generalization ability of iForest, removes redundant iTree to reduce the computational complexity, Detection speed and enhanced stability, improved data anomaly detection performance.

II. ISOLATION FOREST ALGORITHM AND ITS IMPROVEMENT

A. Isolation Forest algorithm

The Isolation Forest algorithm is designed using two features of the anomaly data [11]: a.The exception data accounts for a smaller proportion of the overall size of the data set. b.There is a significant difference between the abnormal data and the attribute value of the normal data. In a training set that contains only numeric types, the data is recursively divided until iTree distinguishes each data from other data. Because they have strong sensitivity to segregation, the anomaly data is closer to the root node of the tree, and the normal data is far from the root node, so that the anomaly data can be detected with a small number of characteristic conditions.

The core of the Isolation Forest algorithm is to build a forest (iForest) consisting of iTree. For ease of description

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and calculation, the Isolation Forest algorithm introduces a definition of the length of the isolation tree.

Definition: Isolation Tree. Let T be a node of an isolation tree. T is either an external-node with no child, or an internal-node with one test and exactly two daughter nodes $(\mathrm{Tl}, \mathrm{Tr})$. Divide the data points into Tl and Tr according to the property values and the size of the split values. To build iTree, randomly select an attribute A and A split value P from the data set $D = \{d_1, d_2, \cdots, d_n\}$, and then divide each data object d_i by the value of its attribute A (called d_i(A)). If $d_i(A) < p$, then it is left in the left subtree and vice versa. In this way, the left and right subtrees are constructed iteratively until one of the following conditions is satisfied: a. there is only one data or several identical data in D; b. the tree reaches its maximum height.

Definition : **Path Length** $h(d)$ of a point d is measured by the number of edges d traverses an iTree from the root node until the traversal is terminated at an external node. Given the data set D, [12] gives the path length of the failed query in the binary search tree:

$$C(n) = 2H(n-1) - (2(n-1)/n) \qquad (1)$$

where $H(i)$ is the harmonic number and it can be estimated by $\ln(i) + 0.5772156649$ (Euler's constant). n is the number of leaves. As $c(n)$ is the average of $h(d)$ given n, we use it to normalise $h(d)$. The anomaly score s of an instance d is defined as:

$$S(d, n) = 2^{\frac{-E(h(d))}{c(n)}} \qquad (2)$$

where $E(h(d))$ is the average of $h(d)$ from a collection of isolation trees.

The Isolation Forest algorithm produces a specified number of iTree and forms iForest. Specifically, the subsets of D are extracted by random sampling to construct each iTree to ensure the diversity of iTree. Treat the data d by traversing the iTree collection in the iForest to determine the leaf node of d. Then, according to the length of its path, the abnormal fraction of d is calculated and the abnormal evaluation of d is carried out.

When $E(h(d)) \to C(n)$, $S \to 0.5$, that is, when all the data is returned to S of 0.5, then there is no obvious abnormal value in all the samples; When $E(h(d)) \to 0$, $S \to 1$, that is, when the S of the data return is very close to 1, then they are outliers; When $E(h(d)) \to n-1$, $S \to 0$, that is, when the S of the data is far less than 0.5, then they have a big potential to be evaluated as normal.

B. Improved Isolation Forest algorithm

Although the Isolation Forest algorithm reduces the obscure and submerged effects to a certain extent and has a low linear time complexity, the artificially introduces stochastic factors, resulting in low accuracy and poor stability. And the construction of iForest, there is a redundant iTree, resulting in a waste of memory space and increase the amount of calculation, affecting the efficiency of the algorithm. In this paper, based on simulated annealing algorithm to improve the iForest forest construction is expected to improve the performance of the algorithm anomaly detection.

Firstly, the difference value between iTree is calculated by Q statistics. D given training set, if the tree T_i can detect d_k, correct the $y_{k,i} = 1$, otherwise $y_{k,i} = 0, i = 1,2,\cdots, L$. The results of the predictions of T_i and T_j are shown in table Ⅰ:

TABLE I. T_i AND T_j TEST RESULTS LISTED IN THE TABLE

	$T_i = 1$	$T_i = 0$
$T_j = 1$	N^{11}	N^{10}
$T_j = 0$	N^{01}	N^{00}

Among them, $n = N^{11} + N^{10} + N^{01} + N^{00}$

The difference between T_i and T_j is $Q_{i,j}$, and the formula is as follows:

$$Q_{i,j} = \frac{N^{11}N^{00} - N^{10}N^{01}}{N^{11}N^{00} + N^{10}N^{01}} \qquad (3)$$

$$Q = \begin{bmatrix} Q_{11} & Q_{12} & \cdots & Q_{1L} \\ Q_{21} & Q_{22} & \cdots & Q_{2L} \\ \vdots & \vdots & & \vdots \\ Q_{L1} & Q_{L2} & \cdots & Q_{LL} \end{bmatrix} \qquad (4)$$

Among them, N^{ab} in T_i and T_j detection D_{Train} have n samples, meet $y_{k,i} = a$ and $y_{k,j} = b$, the number of samples, $k = 1,2,\cdots, n$. Q represents the difference matrix of L iTree. Statistics suggest that if two iTree is independent, the value of the two iTree's Q statisticss is 0. The value of the Q statistics varies between $[-1, 1]$. The greater the value of Q statisticss, the smaller the difference between the two iTree. In extreme cases, if the value of two iTree's Q statisticss is 1, then the two iTree differences are 0.

Then, the accuracy value of each of iTree was calculated by cross-validation. First of all, the original training data is divided into equal amount and mutually disjoint subsets N, $N - 1$ each use a subset of the training, the rest of the test a subset, N copies of data one by one as a test set for training and testing, eventually to N for the measurement of average tree iTree precision value for X. The aim is to: on the premise of guarantee the iTree features the same tree, can more effectively reflect their respective actual detection ability, the result often than in the original training set to test accuracy directly on a more real objective.

Finally, according to the difference and accuracy of iTree, the fitness value of each iTree was calculated, and the simulated annealing algorithm was used to select an excellent iTree from T to form iForest. The fitness function is as follows:

$$F(T_j) = \frac{1}{W_1X_j + W_2Q_{ij}} \qquad (5)$$

Among them, $F(T_j)$ is denoted as the fitness function of T_j. X_j represents the exact value of T_j . W_1, W_2 respectively represents the weight of accuracy and difference.

Simulated annealing algorithm was first applied by Kirkpatrick et al. In the field of combinatorial optimization [13]. Randomly select a fitness value of iTree as the initial solution, and by means of a series of Markov chains generated by decreasing the control parameter t, using a new solution generation device and acceptance criteria to repeat the "generate new solution - calculate the objective function

Poor - to determine whether to accept the new solution - to accept or abandon the new solution ", the current solution to the current iteration, the algorithm is terminated when the output of the optimal iTree. Repeated cycles, from the initial forest selected high detection performance of the tree iTree built into iForest, thus reducing the iForest memory space, reducing the detection of computing, improve the accuracy and efficiency.

III. DATA ANOMALY DETECTION ALGORITHM BASED ON SA-IFOREST

In a data set, [14] can be seen based on the definition of abnormal data, and the abnormal data is only a small part of the data. According to the characteristics of the abnormal data much less percentage, first of all, through the training set randomly selected attributes training more iTree tree, they calculated through cross validation of anomaly detection precision, at the same time, the Q statistics to calculate the difference between iTree, then the accuracy and differences as iTree selection criteria, choose a good iTree to build integration iForest, then the test statistics out the anomaly score test set. The algorithm is described as follows:

Input: D represents data sets; N is the subsampling size; L is the number of iTree.

Step 1: Set the maximum height of iTree, initializing iForest.

Step 2: Build the first iTree.

Step 3: Repeat step 2 to construct the remaining $L-1$ tree to form the initial forest T.

Step 4: The initial forest T was trained with training set D_{Train}, and the value of the value of each iTree was calculated by cross-verification using the Q statistics calculation method.

Step 5: Based on the difference and accuracy of iTree, the simulated annealing algorithm was used to select the optimal value of l tree from the initial forest T.

Step 5 includes the following steps:

Step 5.1: Initialization. The initial temperature t_0 is large enough to set the temperature $t = t_0$, to take the initial solution T_1.

Step 5.2: For current temperature t, repeat step 5.3 - step 5.6.

Step 5.3: A new solution T_2 is generated for the current solution T_1 random perturbation.

Step 5.4:Calculate T_2 increment $df = F(T_2) - F(T_1)$, where $F(T_1)$ is the fitness value for tree T_1.

Step 5.5: If $df < 0$, accept T_2 as the new current solution, namely $T_1 = T_2$; Otherwise in accordance with the rules of Metropolis, calculate T_2 probability p, namely random interval (0,1) uniformly distributed random Numbers on rand, if $p > rand$, also accept T_2 as new current solution, namely $T_1 = T_2$, T_1 or keep the current solution.

Step 5.6: If meet the termination conditions of set, then the output current T_1 for optimal solution, termination conditions are usually off for data processing in the continuous chain of several Metropolis T_2 end are not accepted or is set to end temperature; Otherwise, after the attenuation function attenuation temperature t, return step 4.2. The attenuation function is:

$$t_s = \frac{t_0}{\lg(1+\theta)} (\theta = 1,2,\cdots) \qquad (6)$$

Where t_s is the temperature value of θ step, t_0 is the initial temperature.

Step 5.7: Repeat step 5.3-step 5.6. Select $l(l \le L)$ from the initial forest T to have an excellent adaptive value of iTree into iForest.

Step 6: The test data was calculated to calculate the path length $h(d)$ of each iTree. According to its path length $h(d)$, the abnormal fraction $S(d,n)$ of d is calculated and the abnormal evaluation of d is carried out.

IV. EXPERIMENT AND RESULT ANALYSIS

This section analyzes and contrasts the implementation efficiency, effectiveness and stability of SA-iForest, iForest and LOF. To make the experimental data more convincing, the data sets of Breastw, Diabetes, Unbalanced, Http, Shuttle, Pendigits and Messidor_features were selected from the UCI dataset. Refer to the [15] method for each data set. Select samples belonging to the sparse class as exception objects. Table 2 lists the statistical characteristics of each dataset.

TABLE II. EXPERIMENTAL DATASET FEATURES.

Data set name	cardinal number	Number of attributes	Abnormal points
Breastw	683	9	239
Diabetes	768	9	268
Unbalanced	856	33	12
Http	567497	3	2270
Shuttle	49097	9	3437
Pendigits	6870	17	156
Messidor_features	1151	20	540

The experimental platform is a Windows host equipped with 3.60 GHz CPU and 8.00 GB of ram. The default value of SA-iForest execution parameters is: l = 100, m = 256, and the subsampling size is 256. The reason for this is that when m is set to 256, it usually provides enough detail to help us perform abnormal detection in a large number of data sets.

A. Verification of algorithm validity

In order to verify the accuracy of SA-iForest, the Breastw, Diabetes, Unbalanced, Http, Shuttle, Pendigits and Messidor_features data sets were used as test data sets to perform SA-iForest, Isolation Forest and LOF for accuracy testing. Figure 1, Figure 2 shows the Breast, Http two data sets ROC curve. In order to make the results more objective, for each algorithm were run 10 times, and then take the average. The corresponding AUC values are detected on different data sets as shown in Table 3. Where AUC is the area under the ROC curve and the performance of the algorithm is evaluated by comparing the size of the AUC value [16,17]. AUC value between 0.5-1.0, AUC closer to 1.0, indicating the higher the accuracy of the algorithm.

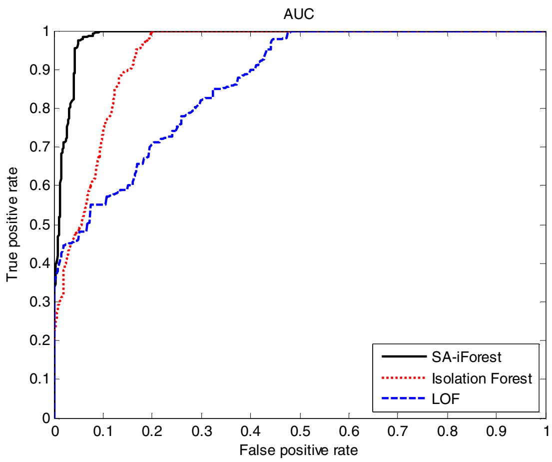

Figure 1. Breastw dataset test results.

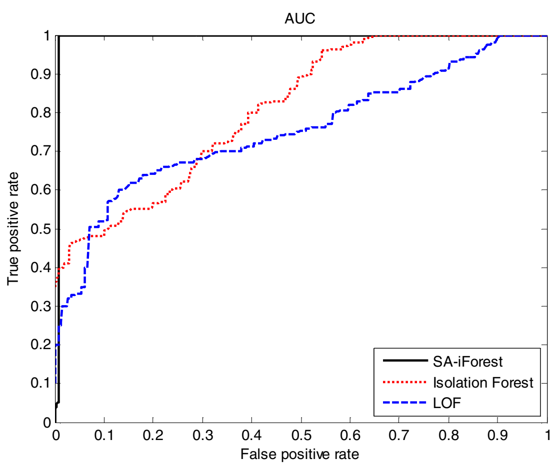

Figure 2. Http dataset test results.

From Figure 1-Figure 2, it can be seen that the SA-iForest algorithm is feasible, and it is better than the iForest algorithm and LOF algorithm. This is mainly because the SA-iForest algorithm each calculated through cross-validation iTree anomaly detection precision, at the same time, the Q statistics to calculate the difference between iTree, using simulated annealing algorithm selected iTree anomaly detection accuracy is relatively high, improve the prediction accuracy of SA-iForest algorithm.

TABLE III. THE CORRESPONDING AUC VALUES ARE DETECTED ON DIFFERENT DATA SETS.

Data set name	SA-iForest	iForest	LOF
Breastw	0.98	0.92	0.84
Diabetes	0.93	0.61	0.55
Unbalanced	0.87	0.57	0.62
Http	0.99	0.89	0.75
Shuttle	1.00	0.73	0.58
Pendigits	0.96	0.77	0.50
Messidor_features	1.00	0.85	0.52

According to table II and table III, the detection accuracy of LOF is higher than that of iForest, and the detection accuracy of SA-iForest is higher than that of LOF. For Http and Shuttle, due to large data sets and abnormal points more, lead to the accuracy of the LOF is relatively low, but because of SA - iForest and iForest direct calculation for the data on the location of the leaf node, abnormal score is obtained by the average path length, finally according to the size of the abnormal scores to evaluate data, whether it is abnormal value. The detection accuracy of SA-iForest and iForest is relatively high, and the AUC value of SA-iForest in both Http and Shuttle data sets is 0.99, 1.00, indicating that SA-iForest is more advantageous in processing big data sets. The detection accuracy of SA-iForest in the data sets of Breastw, Diabetes, Pendigits and Messidor_features is also significantly higher than that of iForest and LOF.

In order to compare the differences between this algorithm and iForest and LOF in stability, the following experiment was conducted on the largest data set Http in table I and ran 20 times in a row, which was shown in figure 3. The running results on other data sets are similar.

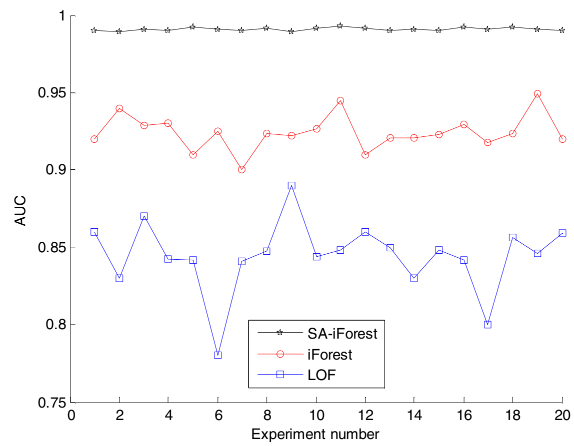

Figure 3. The stability of anomaly detection

As can be seen from Figure 3, the experimental results of the algorithm in the same data set have relatively low fluctuation degree, while the fluctuation of iForest and LOF is relatively large. The standard deviation of SA-iForest, iForest and LOF on the Http data set is $\sigma_1 = 0.0015, \sigma_2 = 0.01148$ and $\sigma_3 = 0.0231$, which shows that the SA-iForest algorithm has better stability.

B. Comparative analysis of execution time

First, the seven data sets in table 1 were used as the test sets, comparing the execution time of SA-iForest, iForest and LOF. In order to make the above three algorithms more objective, the above three algorithms are performed 10 times respectively. Table 4 is the running time for different data sets in SA-iForest, iForest and LOF.

TABLE IV. THE EXECUTION TIME OF DIFFERENT DATA SETS

Data set name	SA-iForest	iForest	LOF
Breast	0.11	0.23	1.14
Diabetes	0.24	0.29	2.07
Unbalanced	0.27	0.38	3.63
Http	8.83	32.67	9186.15
Shuttle	3.21	9.36	736.08
Pendigits	1.40	4.88	376.12
Messidor_features	0.89	1.16	4.54

Can be seen from table □, the execution time of the SA-iForest, iForest significantly below the LOF of execution time, this is because the SA-iForest and iForest no distance or density measurement is used to detect abnormal, so eliminating method based on distance and density of the main distance to calculate the cost. The time complexity of LOF is $O(n^2)$, and the time complexity of Isolation Forest has $O(Lmlog(m))$ in training stage, and the time complexity in the test phase is O (nLlog (m)). $O(L(mlog(m)+1))$, the time complexity of the test phase $O(nllog(m))$ during the training phase. For the Http data set, the total processing time of SA-iForest is 8.83 seconds when the data of $L = 500$, $m = 256$, and $l = 100$ is 567497, the Isolation Forest processing time is 32.67 seconds, which is 3.7 times that of SA-iForest. Indicating that SA-iForest has a lower constant in terms of its computational complexity. Due to SA-iForest algorithm are selected by simulated annealing detection performance better combined into the forest tree, it is not using all the build tree, and iForest algorithm will build all the trees were used for anomaly detection, and compared with iForest algorithm, SA-iForest algorithm deleted some detection performance is poorer trees reduce amount of calculation, and the fast convergence of the simulated annealing optimization algorithm is used to improve the efficiency of network intrusion anomaly detection, so the execution time of the SA-iForest below iForest execution time, but also improve the generalization ability and forecasting performance of the algorithm.

V. CONCLUSION

Aiming at the problem of low detection accuracy, poor execution efficiency and weak generalization ability of Isolation Forest algorithm, this paper presents a SAH-based data anomaly detection method, which is validated by large-scale data set and is carried out with Isolation Forest algorithm and LOF algorithm. In comparison, the experimental results show that SA-iForest is significantly improved in performance efficiency and accuracy and enhances stability. In addition, the algorithm is simple to implement, suitable for dealing with high-dimensional mass data and has a good parallelization potential.

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